AI-enhanced competency transfer hubs: a conceptual framework for university-industry engagement and knowledge sharing

AI 強化型能力轉移樞紐：大學與產業參與及知識共享的概念性框架

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Abstract　摘要

This paper introduces a framework for AI-driven competency transfer hubs, designed to facilitate effective knowledge exchange and collaboration between universities and industries. These hubs leverage artificial intelligence technologies like machine learning and natural language processing to enhance the efficiency and effectiveness of information flows between academic institutions and industry partners, optimizing the whole knowledge-sharing process. Using the TCM-ADO framework the paper consolidates existing perspectives and offers practical suggestions on how to incorporate AI technologies into competency hubs. The discussion further delves into outlining key layers of such hubs including AI-powered knowledge extraction and enrichment, knowledge customization, adaptive project management as well as collaboration outcome enhancement and feedback optimization.

> ▶ **中文譯文**　本文提出一套以 AI 驅動的「能力轉移樞紐（CTHs）」框架，旨在促進大學與產業之間有效的知識交流與協作。這些樞紐運用機器學習（ML）與自然語言處理（NLP）等 AI 技術，提升資訊流動的效率與效果。研究援引 TCM-ADO 框架，整合現有觀點，就如何將 AI 技術融入能力轉移樞紐提供實務建議，並闡述 CTHs 的核心層次：AI 驅動的知識擷取與豐富化、知識客製化、自適應專案管理，以及協作成效提升與回饋優化。

1 Introduction　第一節　緒論

Artificial intelligence (AI) is revolutionizing a wide range of different industries such as healthcare, manufacturing or finance by accelerating innovation and enhancing operational efficiencies. One of AI's most promising transformative roles lies in fostering university-industry collaborations, where academic institutions generate cutting-edge research and industries apply these advancements for real-world impact.

> ▶ **中文譯文**　人工智慧（AI）正以加速創新、提升運營效率的方式，在醫療、製造、金融等各行各業掀起革命。AI 最具前景的轉型角色之一，在於推動大學與產業的協作：學術機構產出前沿研究，產業界則將這些成果應用於實際場景。

University-industry collaborations take various forms, including internships, joint research projects, technology licensing, and consulting agreements. However, traditional university-industry partnerships often struggle with challenges such as misalignment in research and industry timelines, cultural differences, delayed knowledge dissemination and inefficiencies in knowledge transfer. These issues are particularly pronounced in AI-related fields due to the rapid pace of technological advancements, necessitating novel, AI-driven mechanisms to bridge the gap between academia and industry.

> ▶ **中文譯文**　大學與產業的協作形式多元，包括實習、聯合研究計畫、技術授權與顧問委託等。然而，傳統的產學合作普遍面臨挑戰：研究時程與產業需求難以對齊、文化差異、知識傳播遲緩，以及知識轉移效率不足。這些問題在 AI 相關領域尤為突出，技術迭代的速度要求更新穎的 AI 驅動機制，才能彌合學術界與產業界之間的鴻溝。

This study aims to develop a conceptual framework for AI-enhanced competency transfer hubs (CTHs) to enhance knowledge exchange between universities and industries. CTHs are structured, technology-enhanced platforms designed to systematically facilitate knowledge transfer, optimize competency alignment and improve the efficiency of university-industry collaborations. Unlike traditional models such as technology transfer offices or research consortia, AI-enhanced CTHs leverage AI-based technologies like machine learning (ML) or natural language processing (NLP) to streamline knowledge sharing, personalize competency mapping and automate industry-academia matchmaking. By doing so these hubs can significantly improve the scalability, adaptability, and efficiency of knowledge transfer.

> ▶ **中文譯文**　本研究旨在建構 AI 強化型能力轉移樞紐（CTHs）的概念性框架，以提升大學與產業之間的知識交流。CTHs 是有結構、技術強化的平台，旨在系統性地促進知識轉移、優化能力對齊，並提升產學協作效率。有別於傳統的技術移轉辦公室或研究聯盟，AI 強化型 CTHs 透過 ML、NLP 等技術，簡化知識共享流程、客製化能力圖譜，並自動化產學配對。

Despite extensive research on university-industry collaboration models, the role of AI in optimizing knowledge transfer remains significantly underexplored. Prior works such as the Triple Helix model emphasize the importance of government, academia, and industry collaboration, but lack adaptability to the fast-evolving AI landscape. This research is guided by the following research question (RQ): How can AI-enhanced CTHs improve the efficiency, scalability, and adaptability of university-industry knowledge exchange?

> ▶ **中文譯文**　儘管產學合作模式已有大量研究，AI 在優化知識轉移中的角色仍嚴重不足。三螺旋模型強調政府、學術與產業的協作，但缺乏對 AI 快速演進的適應性；野中郁次郎的知識轉移框架與開放創新理論對 AI 如何自動化、規模化知識交流亦未完整回應。本研究核心問題為：AI 強化型 CTHs 如何提升大學與產業知識交流的效率、可擴展性與適應性？

This study argues that AI-driven CTHs serve as a complementary mechanism rather than a replacement of traditional models. By automating knowledge transfer and dynamically aligning research outputs with industry needs, AI-enhanced CTHs facilitate real-time data-driven decision-making to align research with industry priorities, funding opportunities, and commercialization pathways. AI-powered partnerships must also emphasize fairness, transparency and accountability, ensuring that ethical governance remains central to these interactions.

> ▶ **中文譯文**　本研究主張，AI 驅動的 CTHs 是對傳統模式的「補充」而非「取代」，透過自動化知識轉移並動態對齊研究產出與產業需求，促進即時的數據驅動決策，讓研究成果與產業優先事項、資金機會和商業化路徑緊密對齊。AI 驅動的夥伴關係必須重視公平、透明與問責，確保倫理治理始終處於核心位置。

2 Literature Review and Conceptual Grounding 第二節　文獻回顧與概念基礎

2.1 University-Industry Collaboration　2.1　大學與產業的協作

Nowadays, in many countries around the world, one of the main directions of innovation policy refers to the encouragement of more intensive commercialization of university knowledge. Therefore, universities have become an essential part of regional entrepreneurial ecosystems. University-industry collaborations have long been recognized as essential drivers of technological progress, innovation, and economic development. The Triple Helix model underscores the significance of partnerships among universities, industries, and governments in fostering a knowledge-based economy — universities act as knowledge generators, industries serve as knowledge consumers, while governments establish regulatory frameworks. While the Triple Helix model remains influential, it does not fully account for the increasing complexities of knowledge transfer in rapidly evolving technological landscapes.

> ▶ **中文譯文**　當今許多國家的創新政策主軸之一，是鼓勵大學知識更積極地商業化，大學已成為區域創業生態系的核心角色。產學合作長期被視為技術進步、創新與經濟發展的關鍵驅動力。三螺旋模型強調大學（知識生產者）、產業（知識消費者）與政府（監管框架）三方夥伴關係的重要性。然而，三螺旋模型在面對快速演進的技術環境時，對知識轉移日益複雜的現況仍有不足。

Traditional knowledge transfer models have highlighted the importance of structured mechanisms to ensure that academic research translates into practical innovations. However, existing theoretical models such as Open Innovation and Absorptive Capacity Theory are not fully helpful to address how AI can enhance and automate the knowledge transfer process. In contrast, recent empirical studies have demonstrated that AI-driven solutions can significantly enhance knowledge transfer through automating knowledge discovery, facilitating interdisciplinary connections and accelerating the transfer of research insights into practical applications.

> ▶ **中文譯文**　傳統知識轉移模型強調結構化機制對於將學術研究轉化為實際創新的重要性。開放創新理論與吸收能力理論提供了企業如何辨識、吸收並應用外部知識的洞察，但對 AI 如何強化、自動化知識轉移流程的回應仍然不足。相比之下，近期實證研究已顯示，AI 驅動的解決方案可透過自動化知識探索、促進跨學科連結、加速研究洞察轉化，顯著強化產學間的知識轉移。

2.2 Knowledge Transfer and AI　2.2　知識轉移與 AI

Effective knowledge transfer is the cornerstone of successful university-industry collaboration. The Knowledge Transfer Theory (Nonaka & Takeuchi, 1995) emphasizes the systematic codification and exchange of knowledge. However, contemporary research increasingly suggests that knowledge transfer also thrives in decentralized, organic, and individual-driven exchanges, where personal networks, informal collaborations, and departmental initiatives play a significant role.

> ▶ **中文譯文**　有效的知識轉移是成功產學協作的基石。野中郁次郎的知識轉移理論（Nonaka & Takeuchi, 1995）強調知識的系統性編碼與交換。然而，當代研究越來越指出，知識轉移也在去中心化、有機的個人驅動交流中蓬勃發展，個人網絡、非正式協作與部門主導的行動都扮演關鍵角色。

The complexity of knowledge exchange has intensified with the advent of AI. Unlike conventional models that rely on human mediation and structured exchanges, AI introduces adaptive mechanisms capable of enhancing both formal and informal knowledge flows. ML, NLP, and predictive analytics facilitate the automatic identification, extraction, and dissemination of knowledge, transforming static, linear transfer models into dynamic, self-sustaining ecosystems that align research advancements with industry needs in real time. AI-driven CTHs provide a scalable, hybrid solution by integrating institutional oversight with decentralized collaboration; NLP-driven knowledge extraction systematically captures insights from conferences, workshops, and informal collaborations, ensuring tacit knowledge is not lost.

> ▶ **中文譯文** AI 的出現讓產學知識交流的複雜性大幅加劇，打破傳統依賴人工中介的轉移機制。ML、NLP 與預測分析能自動辨識、擷取並傳播知識，將靜態的線性轉移模型轉變為動態、自我維繫的生態系統，讓研究進展與產業需求即時對齊。AI 驅動的 CTHs 透過整合機構監督與去中心化協作，提供混合解決方案；NLP 驅動的知識擷取則系統性地捕捉研討會、工作坊與非正式協作中的洞察，確保隱性知識不流失。

2.3 TCM-ADO Framework 2.3 TCM-ADO 分析框架

The foundation of this study is built upon the TCM-ADO framework (Paul et al., 2017). This framework encompasses two main elements: TCM (Theory, Context and Methodology) and ADO (Antecedents, Decisions, Outcomes). The six dimensions are applied to AI-driven CTHs as follows: Theory (T) — knowledge transfer, open innovation, and socio-technical systems theory; Context (C) — AI-driven collaborative environments and barriers; Methodology (M) — conceptual model building; Antecedents (A) — drivers for needing CTHs; Decisions (D) — choices in adopting AI tools; Outcomes (O) — enhanced efficiency and accelerated innovation cycles.

> ▶ **中文譯文** 本研究以 Paul 等人（2017）提出的 TCM-ADO 框架為基礎，包含 TCM（理論、情境、方法論）與 ADO（前因、決策、結果）兩大元素。應用於 AI 驅動 CTHs 的六個維度為：理論（T）涵蓋知識轉移理論、開放創新與社會技術系統理論；情境（C）聚焦 AI 驅動的協作環境與障礙；方法論（M）為概念模型建構；前因（A）是需要 CTHs 的驅動因素；決策（D）是採用 AI 工具的選擇；結果（O）則是效率提升與創新週期加速。

3 Conceptual Model Development　第三節　概念模型建構

3.1 AI-Driven Competency Transfer and Collaboration Framework　3.1　AI 驅動的能力轉移與協作框架（四層架構）

The new AI-Enhanced Competency Transfer and Collaboration Framework aims to strengthen relationships between universities and industries by streamlining knowledge sharing processes and enhancing collaboration scalability. The setup consists of four layers, working together to facilitate effective knowledge sharing processes.

> ▶ **中文譯文**　這套 AI 強化型能力轉移與協作框架，旨在透過簡化知識共享流程、提升協作可擴展性，強化大學與產業之間的關係。整體架構由四個層次組成，協同運作以促進有效的知識共享流程。

Layer 1: Knowledge Extraction and Enrichment　第一層：知識擷取與豐富化

The initial step involves gathering and processing data to establish a base for further knowledge sharing. By utilizing machine learning techniques for data scraping and aggregation purposes, academic articles along with patents and industry reports are compiled into a database. A Knowledge Graph is created using NLP techniques to link research subjects and prospects. The knowledge enrichment module uses entity recognition to provide context, ensuring knowledge is customized to suit collaboration purposes like product development or research innovation.

> ▶ **中文譯文**　第一層核心工作是蒐集並處理資料，為後續知識共享奠定基礎。透過 ML 技術進行網路爬取與資料彙整，學術期刊文章、專利與產業報告被整合進資料庫。接著，利用 NLP 技術建立知識圖譜，連結研究主題與應用前景。知識豐富化模組藉由實體識別為資訊提供脈絡，確保知識能客製化以符合產品開發或研究創新等協作目的。

Layer 2: Automated Knowledge Transfer and Customization　第二層：自動化知識轉移與客製化

This layer utilizes AI tools to adjust and share information fitting each individual's needs, dealing with concerns regarding terminology clarity and practical usability. NLP-powered summarization simplifies research discoveries for industry professionals. Interactive AI elements respond to inquiries and explain complicated subjects, helping connect the divide between academic and industry knowledge. NLP translation models help align language with industry-specific terminology, fostering coherent communication.

> ▶ **中文譯文**　第二層運用 AI 工具，依據個人需求調整並共享資訊，處理術語清晰度與實際可用性的問題。NLP 驅動的摘要功能將研究發現簡化，讓產業人士能理解。互動式 AI 元素可即時回應詢問、解釋複雜概念，縮短學術與產業知識之間的落差。NLP 翻譯模型則有助於讓語言與產業慣用術語對齊，促進各方的一致溝通。

Layer 3: Scalability, Coordination, and Adaptive Project Management　第三層：可擴展性、協調與自適應專案管理

The third layer assists in managing day-to-day tasks effectively to facilitate responsive teamwork. AI-based analytics for resource planning and schedule adjustment enable better goal alignment. NLP tools help coordinate collaborations across fields of expertise. AI streamlines workflows and provides adaptability and alignment of project schedules, enabling teams to swiftly adjust to emerging insights and market changes.

> ▶ **中文譯文**　第三層協助有效管理日常任務，促進靈活的團隊協作。AI 分析進行資源規劃、依學術與商業時程調整排程，讓組織更好地與目標對齊。NLP 排程與翻譯工具有助於跨領域協作。AI 在

簡化工作流程方面的應用，提供專案時程的適應性與對齊性，讓團隊迅速因應新興洞察與市場變化。

Layer 4: Collaboration Outcome Enhancement and Feedback Optimization　第四層：協作成效提升與回饋優化

The last stage strengthens collaboration results by providing feedback and adaptable methods. Machine learning techniques support impact analysis and monitor collaboration performance indicators. NLP-driven sentiment analysis evaluates communication emotions and team interactions. A knowledge preservation component saves insights and top-notch strategies in an accessible database. Long-term team compatibility measures enhance fit and team formation strategies to boost collaboration efficiency.

▶ **中文譯文**　第四層透過回饋機制與自適應方法強化協作成果。ML 技術支援影響力分析並監測協作績效指標；NLP 驅動的情感分析評估溝通情緒與團隊互動；知識保存模組將洞察與最佳策略儲存入可取用的資料庫。長期團隊相容性評估有助於提升配對精準度與協作效率。

3.2 Defining the Core Elements of an AI-Enhanced Competency Transfer Hub　3.2　AI 強化型能力轉移樞紐的核心構成要素

As detailed in Table 1, the implementation and success of the suggested AI-driven CTHs depend on key variables spanning financial, technological, and regulatory categories. By aligning resources and expertise from both academia and industry, AI-Enhanced CTHs provide a robust framework for converting research insights into scalable, real-world applications.

▶ **中文譯文**　如表 1 所示，AI 驅動 CTHs 的實施與成功，取決於橫跨財務、技術與法規等類別的關鍵變數。透過整合產學雙方的資源與專業知識，AI 強化型 CTHs 提供了將研究洞察轉化為可擴展實際應用的強健框架。

Table 1 Key Variables of AI-Driven Knowledge Hubs　表 1　AI 驅動知識樞紐的關鍵變數

Variable Category 變數類別	Key Variables 關鍵變數
Funding & Financial 資金與財務	Capital investment, operational budget, return on investments 資本投資、營運預算、投資報酬
Technological 技術	High-performance computing, data quality, AI tools, cybersecurity 高效能運算、資料品質、AI 工具、網路安全
Skills Development 技能發展	Skill level, training programs, interdisciplinary readiness 技能水準、訓練計畫、跨學科準備度
Governance 治理	Organizational structure, IP/data sharing, stakeholder engagement 組織架構、智財/資料共享、利害關係人參與
Performance Metrics 績效指標	Project success, innovation output, employment outcomes 專案成功率、創新產出、就業成果
Collaborative Culture 協作文化	Partnership alignment, incentive structures, communication tools 夥伴關係對齊、激勵結構、溝通工具
R&D Scope 研發範疇	Target research areas, scalability, readiness levels 目標研究領域、可擴展性、技術成熟度
Regulatory & Ethical 法規與倫理	Compliance, ethical standards, environmental impact 合規性、倫理標準、環境影響

The key components consist of: (1) Automation and Scalability — AI reduces workload and enables simultaneous management via cloud infrastructure; (2) Knowledge Management and Curation — NLP and ML automate sorting and retrieval; (3) Stakeholder Integration — intelligent matchmaking algorithms connect academia and industry; (4) Personalized Learning and Adaptive Training — AI customizes learning pathways; (5) Ethical Governance and Compliance — AI streamlines compliance monitoring and tackles algorithmic bias; (6) Performance Monitoring and Continuous Improvement — AI-powered feedback loops build upon past project lessons.

> ▶ **中文譯文** 六大核心構成要素：（1）自動化與可擴展性——AI 透過雲端基礎設施降低工作量並同時管理多個協作；（2）知識管理與策展——NLP 和 ML 自動化分類與擷取；（3）利害關係人整合——智慧配對演算法連結產學雙方；（4）個人化學習與自適應訓練——AI 依個人技能水準客製化學習路徑；（5）倫理治理與合規——AI 簡化合規監控（含 GDPR）並應對演算法偏見；（6）績效監測與持續改善——AI 驅動的回饋迴圈確保協作從過去專案中汲取教訓。

3.3 Advanced AI Integration 3.3 進階 AI 整合

Hybrid AI Models 混合 AI 模型

There is a development in AI research involving hybrid AI models that blend symbolic AI with sub-symbolic AI techniques such as deep learning and machine learning. These hybrid models enhance interpretability and efficiency in decision making. They combine logical inference capabilities with ML techniques to recognize intricate patterns in extensive datasets. In competency transfer scenarios, hybrid AI models can fulfil three roles: (1) Semantic Knowledge Mapping — matches research findings with industry requirements more effectively; (2) Dynamic Resource Allocation — forecasts and distributes resources in response to evolving project requirements; (3) Advanced Personalization — enables tailored interactions while ensuring alignment with knowledge transfer goals.

> ▶ **中文譯文** AI 研究的重要進展是混合 AI 模型，融合符號 AI 與深度學習、ML 等次符號技術，提升決策的可解釋性與效率。這類模型同時運用邏輯推理與 ML 技術辨識複雜資料模式，在能力轉移場景中扮演三種關鍵角色：（1）語意知識映射——更有效地將研究發現與產業需求配對；（2）動態資源分配——預測並因應專案需求演變調配資源；（3）進階個人化——在確保知識轉移目標對齊的前提下，實現產學雙方的客製化互動。

Explainability and Interpretability in AI (XAI) 可解釋 AI（XAI）

The concept of Explainable AI (XAI) has been getting recognition because it allows for understandable decision-making processes. Integrating XAI into the AI-enhanced CTHs framework ensures that both academic and industry collaborators can rely on and comprehend AI system decisions. XAI improves trust, provides domain-specific explanations (e.g., healthcare, cybersecurity contexts), and enables feedback loops where users refine AI recommendations over time — enhancing transparency and adaptability.

> ▶ **中文譯文** 可解釋 AI（XAI）的概念正獲得越來越多認可，因為它讓決策過程變得可理解。將 XAI 整合進 CTHs 框架，確保產學雙方都能信賴並理解 AI 系統的決策。XAI 提升對 AI 決策的信任、提供領域專屬解釋（如醫療、網路安全情境），並建立讓使用者能持續優化 AI 建議的回饋迴圈，強化透明度與適應性。

Ethical AI Considerations 倫理 AI 考量

The integration of AI-driven CTHs presents critical ethical challenges. A major concern is algorithmic bias in AI-powered recommendation systems, which influence partner selection, funding allocation and knowledge exchange. If AI models are trained on historically skewed datasets, they may reinforce existing academic hierarchies, favoring well-established institutions while marginalizing emerging disciplines and early-career researchers. To mitigate this, bias detection must be embedded within AI models, XAI frameworks must provide interpretable reasoning, and human-in-the-loop oversight should allow academic experts to review AI-generated suggestions. Data privacy must be addressed through differential privacy and federated learning architectures.

> ▶ **中文譯文** AI 驅動 CTHs 的整合帶來重大倫理挑戰。最核心的憂慮是 AI 推薦系統中的演算法偏見，影響夥伴選擇、資金分配與知識交流。若 AI 模型以歷史上有偏差的資料集訓練，可能強化現有學術階層，使成熟機構受益，同時邊緣化新興學科與早期研究者。為降低此風險，偏見偵測應內建於 AI 模型；XAI 框架應提供可解釋的推理；人機協作的監督機制應允許學術專家審視 AI 建議。資料隱私則應透過差異隱私與聯邦學習架構加以保護。

4 Discussion and Key Implications 第四節 討論與主要啟示

The use of AI models for the development of AI-enhanced CTHs brings an innovative approach to university-industry partnerships. These models greatly improve the understanding of information and decision-making processes within collaborative environments. Applying the TCM-ADO framework: Theory (T) — hybrid AI expands knowledge transfer theory by linking research to industry requirements; Context (C) — in fast-evolving fields AI enables real-time information sharing and adaptability; Methodology (M) — the framework combines ML with reasoning to ensure partnerships are effective and customized; Antecedents (A) — hybrid AI simplifies information organization and resource distribution; Decisions (D) — universities and industries opt for NLP or ML to streamline knowledge acquisition; Outcomes (O) — enhance effectiveness and expandability in sharing knowledge, boosting innovation timelines.

> ▶ **中文譯文** 透過 TCM-ADO 框架：理論（T）上，混合 AI 模型融合符號 AI 與 ML，擴展了知識轉移理論；情境（C）上，AI 在快速演進領域促進即時資訊共享；方法論（M）上，框架透過系統性審查確保夥伴關係有效且客製化；前因（A）上，混合 AI 系統透過簡化資訊組織與強化資源分配解決核心問題；決策（D）上，產學雙方選擇 NLP 或 ML 工具以簡化知識取得；結果（O）上，重點在於強化知識共享程序的效能與可擴展性，並加速創新時程。

The practical impacts of AI-powered CTHs extend broadly. For universities, the primary advantage lies in the commercialization of research outcomes facilitated by AI which matches scholarly research results with industry demands instantly. For industries, AI-powered CTHs help to cut down costs associated with collaborations and make academic resources more accessible to smaller companies, promoting innovation. Policymakers can leverage AI-enhanced CTHs to aid national innovation strategies, ensuring research funding supports overarching economic goals such as renewable energy and healthcare advancements.

> ▶ **中文譯文** AI 驅動 CTHs 的實際影響廣泛。對大學而言，核心優勢在於 AI 技術能即時將研究成果與產業需求配對，加速研究商業化；對產業而言，CTHs 有助於降低協作成本，讓中小企業也能取得學術資源；對政策制定者而言，AI 強化型 CTHs 可支援國家創新策略，確保研究資金支持整體經濟目標，如再生能源與醫療保健進步。

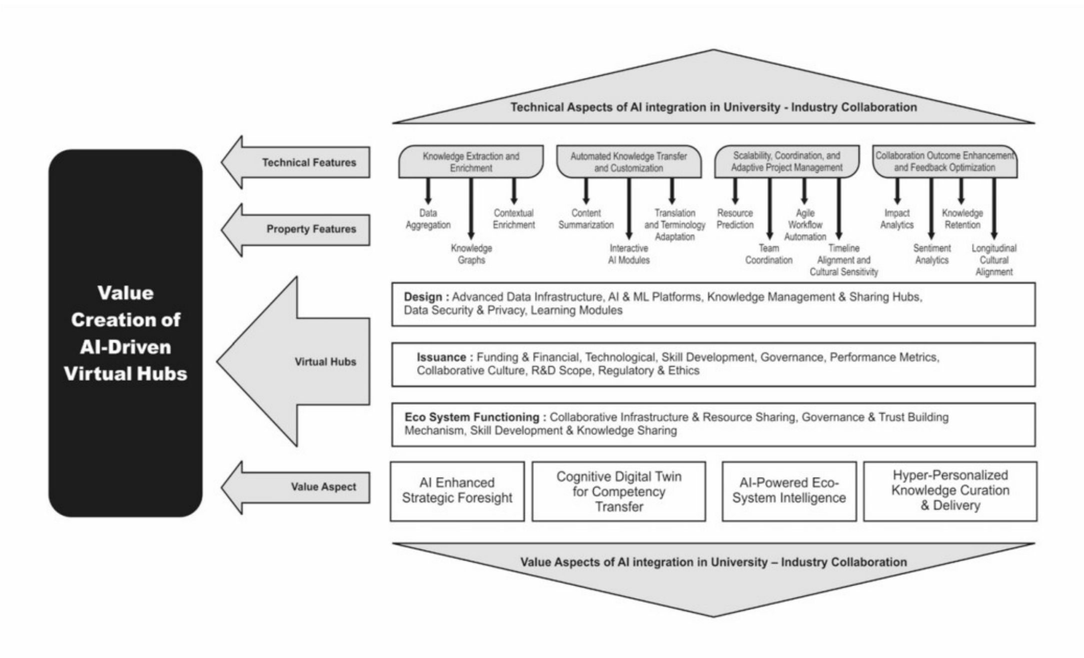

Fig. 1 Analysis Framework

Fig. 1 Analysis Framework Source: Authors 圖１　分析框架（AI 驅動虛擬樞紐的價值創造體系）

5 Conclusion & Future Directions 第五節 結論與未來研究方向

The practical implications of AI-driven CTHs for universities, businesses and policymakers are extensive. These hubs provide a method for collaboration between universities and businesses facilitating knowledge exchange, accelerating innovation processes and increasing societal influence. By utilizing AI advancements like machine learning and natural language processing, universities and businesses can break through obstacles to collaboration and establish more flexible and adaptable innovation environments. Universities benefit from simplified research commercialization; industries benefit from lower costs and research access; policymakers can promote national innovation agendas and monitor collaboration effectiveness.

> ▶ **中文譯文** AI 驅動 CTHs 對大學、企業與政策制定者的實際影響廣泛。這些樞紐提供了產學協作的新途徑，促進知識交流、加速創新進程並擴大社會影響力。透過 ML 與 NLP 等 AI 技術，大學與企業得以突破協作障礙，建立更靈活、更具適應性的創新環境。大學受益於更簡化的研究成果商業化流程；產業則發現這些樞紐有助於降低合作成本並取得最新研究資源；政策制定者則可藉此推動國家創新議程並監測協作成效。

One of the limitations of the current study is that its main emphasis has been on machine learning and natural language processing as AI technologies for CTHs. However, upcoming investigations might probe into other AI-based tools like computer vision or blockchain. Further studies could also investigate how AI-improved CTHs influence innovation results in the long run. Research could look into how AI-powered CTHs can be managed to maintain openness and responsibility — especially concerning data privacy, biases and the need for transparency — particularly when AI-driven choices hold economic consequences.

> ▶ **中文譯文** 本研究的主要限制在於聚焦 ML 與 NLP 兩種 AI 技術，未來研究可探索電腦視覺或區塊鏈等其他 AI 工具的潛力。此外，未來研究可深入探討 AI 強化型 CTHs 對長期創新成果的影響，以及如何在 AI 驅動決策具有經濟後果的情況下，有效管理 CTHs 以維持透明度與問責性——特別是在資料隱私、偏見與公平性等倫理議題上。